Week 8
GUI Programming using AWT
Program 22
Write a Java program to create a simple AWT window containing Label, TextField, and Button.

```
import java.awt.*;

class AWTWindow {
   public static void main(String[] args) {
      Frame f = new Frame("Simple AWT Window");

      Label l = new Label("Enter Name:");
      l.setBounds(50, 80, 100, 30);

      TextField tf = new TextField();
      tf.setBounds(160, 80, 150, 30);

      Button b = new Button("Submit");
      b.setBounds(120, 130, 80, 30);

      f.add(l);
      f.add(tf);
      f.add(b);

      f.setSize(400, 250);
      f.setLayout(null);
      f.setVisible(true);
   }
}
```

OUTPUT:
A window titled Simple AWT Window appears containing a Label (Enter Name:), a TextField, and a Button (Submit).

Program 23
Write a Java program demonstrating different layout managers (FlowLayout and GridLayout).

```
import java.awt.*;

class LayoutDemo {
   public static void main(String[] args) {
      Frame f = new Frame("Layout Managers");

      f.setLayout(new GridLayout(2, 1));

      Panel p1 = new Panel();
      p1.setLayout(new FlowLayout());

      p1.add(new Button("One"));
```

```
        p1.add(new Button("Two"));
        p1.add(new Button("Three"));

        Panel p2 = new Panel();
        p2.setLayout(new GridLayout(2, 2));

        p2.add(new Button("A"));
        p2.add(new Button("B"));
        p2.add(new Button("C"));
        p2.add(new Button("D"));

        f.add(p1);
        f.add(p2);

        f.setSize(400, 300);
        f.setVisible(true);
    }
}
```

OUTPUT:
A window appears with two sections:
- FlowLayout: Buttons One, Two, and Three are arranged horizontally.
- GridLayout: Buttons A, B, C, and D are arranged in two rows and two columns.

Program 24
Write a Java program to perform addition of two numbers using AWT event handling.

```
import java.awt.*;
import java.awt.event.*;

class AdditionAWT extends Frame implements ActionListener {
    TextField t1, t2, t3;
    Button b;

    AdditionAWT() {
        setTitle("Addition of Two Numbers");
        setLayout(new FlowLayout());

        add(new Label("First Number:"));
        t1 = new TextField(10);
        add(t1);

        add(new Label("Second Number:"));
        t2 = new TextField(10);
        add(t2);

        b = new Button("Add");
        add(b);
```

```
        add(new Label("Result:"));
        t3 = new TextField(10);
        t3.setEditable(false);
        add(t3);

        b.addActionListener(this);

        setSize(300, 200);
        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        int a = Integer.parseInt(t1.getText());
        int b = Integer.parseInt(t2.getText());

        t3.setText(String.valueOf(a + b));
    }

    public static void main(String[] args) {
        new AdditionAWT();
    }
}
```

OUTPUT:

Initially, a window appears with two input fields, an Add button, and a result field.